Fiscal Year:	FY 2011	**Task Last Updated:**	FY 11/03/2010
PI Name:	Nelson, Gregory A. Ph.D.		
Project Title:	NSCOR: Charged Particle Radiation and Resultant Oxidative Stress Elicit Deleterious Functional Changes in the Central Nervous System		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	grnelson@llu.edu	**Fax:**	FY 909-558-4035
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8364
Organization Name:	Loma Linda University		
PI Address 1:	Basic Sciences, Div. Radiation Research 11175 Campus Street		
PI Address 2:	Chan Shun Pavilion, Room A-1024		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-1700	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSCOR Space Radiation NNJ08ZSA003N
Start Date:	01/01/2010	**End Date:**	12/31/2014
No. of Post Docs:	6	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Soltesz, Ivan (University of California, Irvine) Wodarz, Dominik (University of California, Irvine) Limoli, Charles (University of California, Irvine) Vlkolinsky, Roman (Loma Linda University) Obenaus, Andre (Loma Linda University) Fike, John (University of California, San Francisco) Komarova, Natalia (University of California, Irvine) Spigelman, Igor (UCLA School of Dentistry) Raber, Jacob (Oregon Health & Science University)		
Grant/Contract No.:	NNX10AD59G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed project will investigate responses of the mouse brain to several radiation exposure regimes corresponding to galactic cosmic rays and solar particle events. The project will characterize behavioral, electrophysiological, biochemical and structural changes both in vivo and in vitro. Radiation-elicited oxidative stress mechanisms will be investigated as important mediators of the biological responses in both wild type and mutant animals. In addition to characterization of normal tissue responses, the potential enhancement of three late risks by radiation exposure will be explored. These are seizures, ischemic injury and traumatic injury. Mathematical models of the neurotoxicity responses will developed for use in estimating risks and fitted to project-specific data and previously-published observations. The aims of the proposal will address HRP risk 1.2 and at least five of the specific gap questions identified by NASA: CNS-1, -3, -5, -6 and -11.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the responses of the central nervous system to charged particle radiation has two main benefits – risk assessment and optimization of radiotherapy. The general population and aviation crews are exposed to charged particles during cross country and trans-oceanic air travel at high altitudes as well as to radon in their homes and workplaces. While these exposures are not as high as those of astronauts they are not insignificant. Cancer risks have traditionally been the focus of health risk estimates but the new awareness of sensitive targets in the brain such as neurogenic centers in the hippocampus raises question about the negative effects of such radiation on late neurodegenerative diseases and potential decline of cognitive abilities. This project directly addresses the cognitive and neurogenesis issues and indirectly addresses late neurodegenerative effects which share reactive oxygen chemistry with radiation as underlying mechanisms. Alzheimer's, Huntington's and in particular, ALS diseases are of interest. In the last ten years, the use of charged particles in radiation therapy has grown dramatically with proton and carbon ion treatment centers being established in the U.S., Europe and Asia. Always limiting the effectiveness of radiotherapy is the response of normal tissue near the tumor site. The precise dose localization associated with charged particle therapy makes it ideal for tumors of the head and neck where careful avoidance of critical structures such as the optic nerve is critical. Knowledge of the dose "tolerance" of brain tissue and the reactions to relatively low doses is critical in optimizing these modalities of treatment.
Task Progress:	The official period of performance for the project is from January 1, 2010 for five years. The final award documentation was signed Jan. 28, 2010 which authorized Loma Linda University (LLU) to draw initial funds and commence activities. Pre-award planning meetings were held with all co-investigators at LLU in November 2009 to expedite subcontract implementation and the final signatures for subcontracts to collaborative institutions were signed either March 22 (UCLA, OHSU) or April 16 (UCI, UCSF), which enabled the collaborators to begin technical work around April 1 or May 1, 2010. With subcontracts in place, initial animal orders were placed and the first proton irradiations of animals in support of Aim 1 were conducted on June 4, 2010. Biological endpoints are all measured at 90 days post irradiation so results from the first measurements on irradiated animals became available around September 4. Subsequent irradiations of wild type and the first cohort of catalase-overexpressing MCAT transgenic animals were performed on July 30 and October 11, respectively. Measurements on these animals will be performed starting on October 30, 2010 and Jan 11, 2011. Preliminary data sets are available for some behavioral and electrophysiological experiments. In parallel to the irradiation experiments, equipment upgrades for electrophysiology at LLU and UCLA were implemented and experimental workstations were tuned for the planned experiments. Baseline measurements were taken from unirradiated control animals. New staff members were also recruited, hired and trained at the participating institutions. Beam time proposals and animal care and use protocols have been approved to support the work. In vitro experiments began immediately after funding and cells were irradiated at Loma Linda. The UCI team is conducting the first heavy ion experiments from November 1 to 22. A kick-off meeting for modeling activities was held at UCI in January with outside participants Sylvain Costes, Ph.D. (Lawrence Berkeley National Laboratory) and Philip Hahnfeldt, Ph.D. (Tufts University) who oversee similar activities for two other NASA NSCOR projects. Finally, the members of the scientific advisory committee were notified of the award and the chairman, Martin Hauer-Jensen, M.D., Ph.D. (U. Arkansas Medical Center) was briefed on progress by the NSCOR director and associate director at the recent Radiation Research Society Meeting. Initial findings: Electrophysiology studies show proton-induced alterations in CA1, CA3 and DG regions of the hippocampus using MED64 and patch clamp methods and the throughput rate for analysis will be greatly enhanced over the previous NSCOR. This will improve statistical power for the analysis which is necessary for low dose measurements. Behavioral testing shows that contextual fear conditioning is sensitive for doses of protons below 1 Gray. Immunohistochemical analysis of neurogenesis is in progress using tissues from animals that had undergone behavioral training. In vitro experiments with protons indicate that there are cyclical changes in the redox status of both mouse and human neural precursor cells. The responses are also sensitive to dose rate and adaptation produced by low priming exposures prior to higher challenge doses.
Bibliography Type:	Description: (Last Updated: 03/26/2026)
Abstracts for Journals and Proceedings	Fike JR, Rosi S, Raber J. "Alterations in cognition and expression of the behaviorally-induced immediate early gene arc due to traumatic brain injury are ameliorated by a previous exposure to 56Fe." 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, New York, May 16-19, 2010. Abstract Book. 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, New York, May 16-19, 2010. , May-2010
Abstracts for Journals and Proceedings	Limoli C, Nelson G, Rabin B, Eisch A, Fike J, O'Banion MK. "CNS Risks: Red and Blue Team Debate." 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, New York, May 16-19, 2010. Abstract Book. 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, New York, May 16-19, 2010. , May-2010
Abstracts for Journals and Proceedings	Nelson GA. "Low Dose Particle Effects on the Brain." 13th Workshop on Ion Beams in Biology and Medicine, Berkeley, California, October 26 – 29, 2010. Abstract and invited talk. 13th Workshop on Ion Beams in Biology and Medicine, Berkeley, California, October 26 – 29, 2010. , Oct-2010

Abstracts for Journals and Proceedings	Nelson, GA, Chang, P, Favre C, Fike JR, Komorova N, Limoli C, Mao XW, Obenaus A, Raber J, Spigelman I, Soltesz I, Song SK, Stampanoni M, Vlkolinsky R, Wodarz D "NSCOR Transition: From Progressive Alterations of CNS Structure & Function are Caused by Charged Particle Radiation to Charged Particle Radiation & Resultant Oxidative Stress Elicit Deleterious Functional Changes in the CNS" 2010 NASA Human Research Program Investigator's Workshop, Feb. 2-5, 2010. Houston, TX Abstract Book , Feb-2010
Abstracts for Journals and Proceedings	Nelson GA, Chang P, Favre C, Fike J, Komorova N, Limoli C, Mao XW, Obenaus A, Raber J, Spigelman I, Soltesz I, Song SK, Stampanoni M, Vlkolinsky R, Wodarz D. "CNS NSCOR in Transition: Charged Particle Radiation & Resultant Oxidative Stress Elicit Deleterious Functional Changes in the CNS." 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, New York, May 16-19, 2010. Abstract Book. 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, New York, May 16-19, 2010. , May-2010
Abstracts for Journals and Proceedings	Rosi S, Ferguson R, Fishman K, Levy W, Raber J, Obenaus A, Fike JR. "Trauma Induced Alterations in Cognition and Expression of the Behaviorally-Induced Immediate Early Gene Arc Are Reduced by a Previous Exposure to 56Fe." 2010 NASA Human Research Program Investigators' Workshop, Houston, TX, Feb. 2-5, 2010. Abstract book. 2010 NASA Human Research Program Investigators' Workshop, Houston, TX, Feb. 2-5, 2010. , Feb-2010
Abstracts for Journals and Proceedings	Tseng B, Tran K, Lan M, Giedzinski E, Limoli C. "Low dose charged particle effects in rodent and human neural stem cells." 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, New York, May 16-19, 2010. Abstract Book. 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, New York, May 16-19, 2010. , May-2010
Abstracts for Journals and Proceedings	Villasana L, Raber J. "Effects of irradiation and the antioxidant, alpha-lipoic acid, on cognitive function in C57BL/6J mice." 2010 Student Research Forum, Oregon Health and Sciences University, May 13-14, 2010. Symposium. 2010 Student Research Forum, Oregon Health and Sciences University, May 13-14, 2010. , May-2010
Abstracts for Journals and Proceedings	Villasana L, Raber J. "Effects of irradiation and the antioxidant, alpha-lipoic acid, on cognitive function in C57BL/6J mice." The American Aging Association's 39th Annual Meeting and 24th Annual Meeting of the American College of Clinical Gerontology, Portland, Oregon, June 2010. Proceedings. The American Aging Association's 39th Annual Meeting and 24th Annual Meeting of the American College of Clinical Gerontology, Portland, Oregon, June 2010. , Jun-2010
Abstracts for Journals and Proceedings	Villasana L, van Meer P, Raber J. "Age- and apoE isoform-dependent effects of 56Fe irradiation on spatial memory retention in the water maze task." 2010 Oregon Chapter Society for Neuroscience, McMinnville, Oregon, March 26-27, 2010. Proceedings. 2010 Oregon Chapter Society for Neuroscience, McMinnville, Oregon, March 26-27, 2010. , Mar-2010
Articles in Peer-reviewed Journals	Acharya MM, Lan ML, Kan VH, Patel NH, Giedzinski E, Tseng BP, Limoli CL. "Consequences of ionizing radiation-induced damage in human neural stem cells." Free Radic Biol Med. 2010 Sep 6. [Epub ahead of print] PMID: 20826207 , Sep-2010
Articles in Peer-reviewed Journals	Raber J. "Unintended effects of cranial irradiation on cognitive function." Toxicologic Pathology 2010;38(1):198-202. PMID: 19880825 , Jan-2010
Articles in Peer-reviewed Journals	Villasana L, Rosenberg J, Raber J. "Sex-dependent effects of 56Fe irradiation on contextual fear conditioning." Hippocampus, 2010 Jan;20(1):19-23. http://dx.doi.org/10.1002/hipo.20659 , Jan-2010
Articles in Peer-reviewed Journals	Villasana L, Pfankuch T, Raber J. "Isoform-dependent effects of apoE on doublecortin-positive cells and microtubule-associated protein 2 immunoreactivity following (137)Cs irradiation." Radiation and Environmental Biophysics, 2010 Aug;49(3):421-6. PMID: 20458592 , Aug-2010
Books/Book Chapters	Obenaus A, Raber J. "Ch 7. Behavioral and neurophysiological changes with exposure to ionizing radiation." in "Military Medicine Consequences of Nuclear Warfare (Revision to 1989 edition)." Ed. R. Zajtchuk, T.J. Cerveny, R.J. Walker. Washington, D.C. : Dept of the Army, 2010. ISBN-13 # 978-0160591341, Dec-2010